

ADVANCED

Stage 6

Trigonometry (Adv), T3 Trig Functions and Graphs (Adv) Trig Graphs (Y12)

Teacher: SHS Maths

Exam Equivalent Time: 87 minutes (based on allocation of 1.5 minutes per mark)



T3 Trig Functions and Graphs



***SmarterMaths analytics based on the average contribution to new syllabus Advanced Maths exams since 2020.**

HISTORICAL CONTRIBUTION

- T3 Trig Functions and Graphs* has contributed a meaningful 5.8% per new syllabus Adv exam since being introduced in 2020.
- This topic has been split into two sub-topics for analysis purposes: *1-Trig Graphs (1.8%), and 2-Trig Applications (4.0%)*.
- This analysis looks at the sub-topic *Trig Graphs*.

HSC ANALYSIS - What to expect and common pitfalls

- Trig Graphs (1.8%)* have been examined in the first three new syllabus exams between 2020-22 and notably omitted in 2023.
- The database size reflects our view that this area will be consistently examined going forward and numerous examples of all trig graph variations are covered.
- We recommend close revision of *T3 EQ-Bank 3 and 5* which are informed by the question style and difficulty level of NESA's sample questions. Also, special attention should be given to *2013 HSC 6 MC* which was surprisingly poorly answered.
- Note that more than half of students answered the 2016 multiple choice question on a tan function graph's *period* incorrectly. Deserves attention.

Questions

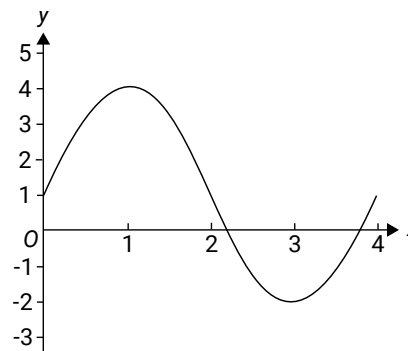
1. Trigonometry, 2ADV T3 2020 HSC 6 MC

Which interval gives the range of the function $y = 5 + 2\cos 3x$?

- A. [2, 8]
- B. [3, 7]
- C. [4, 6]
- D. [5, 9]

2. Trigonometry, 2ADV T3 SM-Bank 8 MC

The diagram below shows one cycle of a circular function.



The amplitude and period of this function are respectively

- A. 3 and 2
- B. 3 and $\frac{\pi}{2}$
- C. 4 and $\frac{\pi}{4}$
- D. 3 and 4

3. Trigonometry, 2ADV T3 SM-Bank 1 MC

$$f(x) = 2\sin(3x) - 3$$

The period and range of this function are respectively

- A. period = $\frac{2\pi}{3}$ and range = $[-5, -1]$
- B. period = $\frac{2\pi}{3}$ and range = $[-2, 2]$
- C. period = $\frac{\pi}{3}$ and range = $[-1, 5]$
- D. period = 3π and range = $[-1, 5]$

4. Trigonometry, 2ADV T3 SM-Bank 2 MC

Let $f(x) = 1 - 2\cos\left(\frac{\pi x}{2}\right)$.

The period and range of this function are respectively

- A. 4 and $[-2, 2]$
- B. 4 and $[-1, 3]$
- C. 1 and $[-1, 3]$
- D. 4π and $[-2, 2]$

5. Trigonometry, 2ADV T3 SM-Bank 3 MC

Let $f(x) = 5\sin(2x) - 1$.

The period and range of this function are respectively

- A. π and $[-1, 4]$
- B. 2π and $[-1, 5]$
- C. π and $[-6, 4]$
- D. 2π and $[-6, 4]$

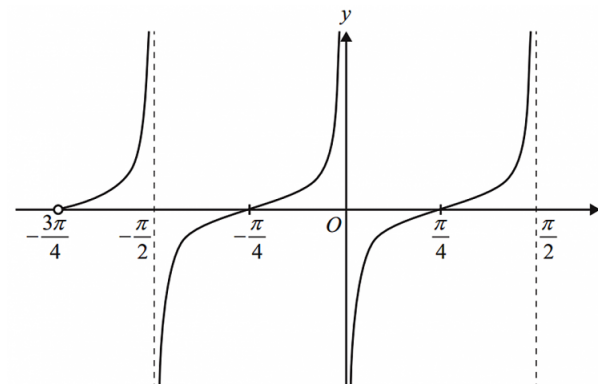
6. Trigonometry, 2ADV T3 SM-Bank 18 MC

The period of the function $f(x) = \tan\left(\frac{\pi x}{2}\right)$ is

- A. 2
- B. 4
- C. 2π
- D. 4π

7. Trigonometry, 2ADV T3 SM-Bank 4 MC

A section of the graph of $f(x)$ is shown below.



The equation of $f(x)$ could be

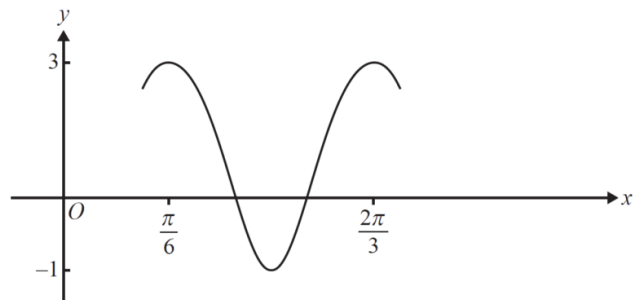
- A. $f(x) = \tan(x)$
- B. $f(x) = \tan\left(x - \frac{\pi}{4}\right)$
- C. $f(x) = \tan\left(2\left(x - \frac{\pi}{4}\right)\right)$
- D. $f(x) = \tan\left(2\left(x - \frac{\pi}{2}\right)\right)$

8. Trigonometry, 2ADV T3 SM-Bank 6 MC

The function with equation $f(x) = 4\tan\left(\frac{x}{3}\right)$ has period

- A. $\frac{2\pi}{3}$
- B. 6π
- C. 3
- D. 3π

9. Trigonometry, 2ADV T3 SM-Bank 7 MC

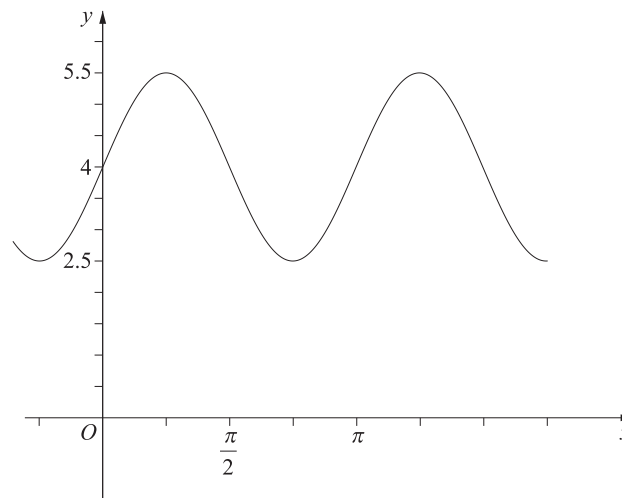


The graph shown could have equation

- A. $y = 2\cos\left(x + \frac{\pi}{6}\right) + 1$
 - B. $y = 2\cos 4\left(x - \frac{\pi}{6}\right) + 1$
 - C. $y = 4\sin 2\left(x - \frac{\pi}{12}\right) - 1$
 - D. $y = 3\cos\left(2x + \frac{\pi}{6}\right) - 1$
-

10. Trigonometry, 2ADV T3 2019 HSC 7 MC

The diagram shows part of the graph of $y = a\sin(bx) + 4$.



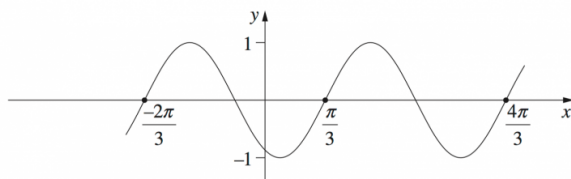
What are the values of a and b ?

- A. $a = 3$ $b = \frac{1}{2}$
 - B. $a = 3$ $b = 2$
 - C. $a = 1.5$ $b = \frac{1}{2}$
 - D. $a = 1.5$ $b = 2$
-

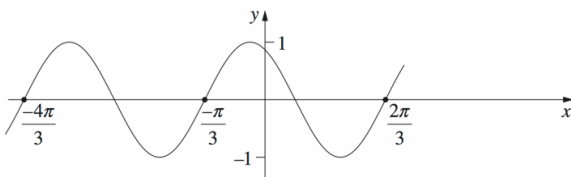
11. Trigonometry, 2ADV T3 2013 HSC 6 MC

Which diagram shows the graph $y = \sin\left(2x + \frac{\pi}{3}\right)$?

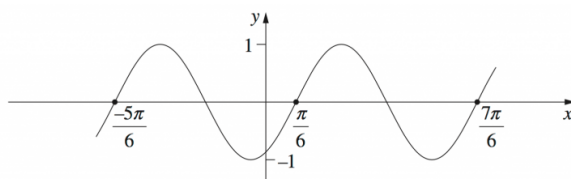
(A)



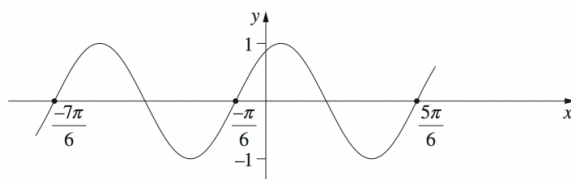
(B)



(C)



(D)



12. Trigonometry, 2ADV T3 2016 HSC 6 MC

What is the period of the function $f(x) = \tan(3x)$?

- A. $\frac{\pi}{3}$
- B. $\frac{2\pi}{3}$
- C. 3π
- D. 6π

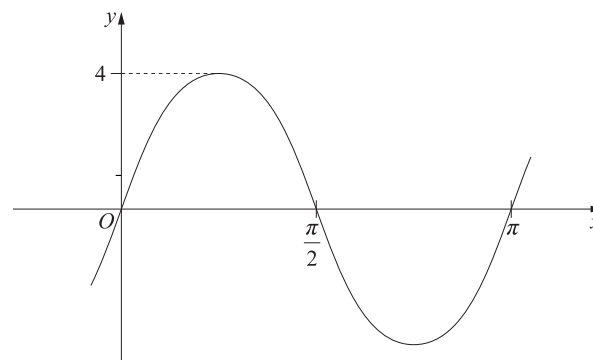
13. Trigonometry, 2ADV T3 EQ-Bank 5

The function $f(x) = \sin x$ is transformed into the function $g(x) = \frac{\sin(4x)}{3}$.

Describe in words how the amplitude and period have changed in this transformation. (2 marks)

14. Trigonometry, 2ADV T3 2010 HSC 8c

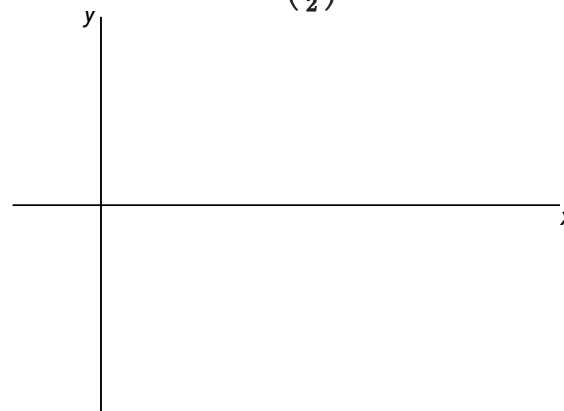
The graph shown is $y = A \sin bx$.



- i. Write down the value of A . (1 mark)
- ii. Find the value of b . (1 mark)
- iii. On the same set of axes, draw the graph $y = 3 \sin x + 1$ for $0 \leq x \leq \pi$. (2 marks)

15. Trigonometry, 2ADV T3 EQ-Bank 6

- a. Sketch the function $y = \cos\left(\frac{x}{2}\right)$ from $0 \leq x \leq 2\pi$ (1 mark)

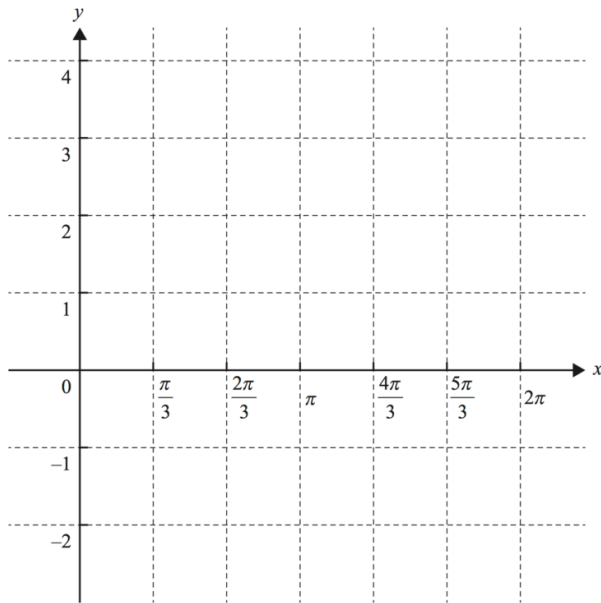


- b. Find the value of x when $y = 0.5$, for $0 \leq x \leq 2\pi$ (2 marks)

16. Trigonometry, 2ADV T3 SM-Bank 9

Let $f(x) = 2\cos(x) + 1$ for $0 \leq x \leq 2\pi$.

- Solve the equation $2\cos(x) + 1 = 0$ for $0 \leq x \leq 2\pi$. (2 marks)
- Sketch the graph of the function $f(x)$ on the axes below. Label the endpoints and local minimum point with their coordinates. (3 marks)



17. Trigonometry, 2ADV T3 2006 HSC 7b

A function $f(x)$ is defined by $f(x) = 1 + 2\cos x$.

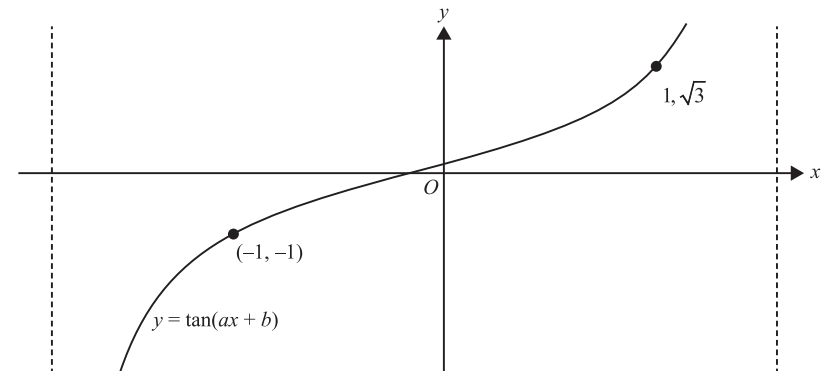
- Show that the graph of $y = f(x)$ cuts the x -axis at $x = \frac{2\pi}{3}$. (1 mark)
- Sketch the graph of $y = f(x)$ for $-\pi \leq x \leq \pi$ showing where the graph cuts each of the axes. (3 marks)
- Find the area under the curve $y = f(x)$ between $x = -\frac{\pi}{2}$ and $x = \frac{2\pi}{3}$. (3 marks)

18. Trigonometry, 2ADV T3 2017 HSC 14a

Sketch the curve $y = 4 + 3\sin 2x$ for $0 \leq x \leq 2\pi$. (3 marks)

19. Trigonometry, 2ADV T3 2010 MET1 3

Shown below is part of the graph of a period of the function of the form $y = \tan(ax + b)$.



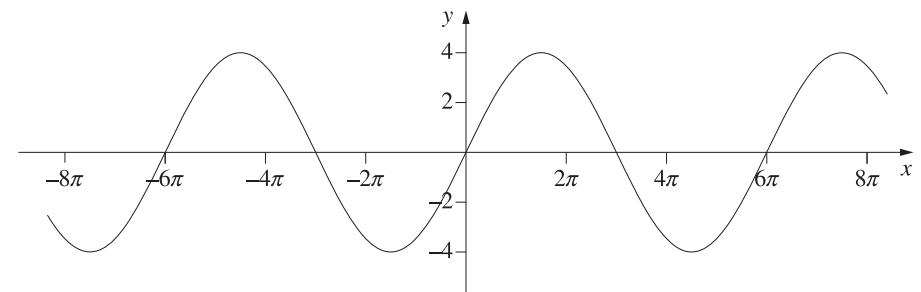
Find the value of a and the value of b , where $a > 0$ and $0 < b < 1$. (3 marks)

20. Trigonometry, 2ADV T3 2021 HSC 20

For what values of x , in the interval $0 \leq x \leq \frac{\pi}{4}$, does the line $y = 1$ intersect the graph of $y = 2\sin 4x$? (2 marks)

21. Trigonometry, 2ADV T3 2022 HSC 14

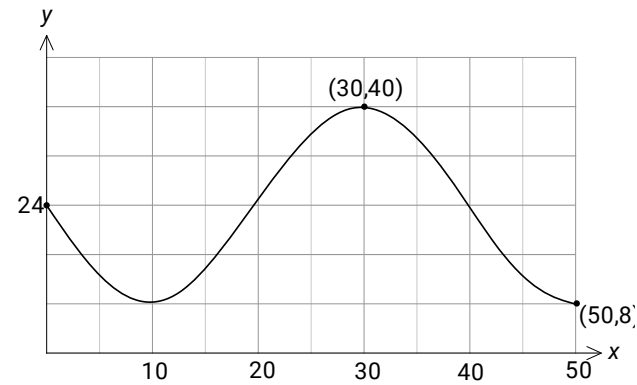
The graph of $y = k\sin(ax)$



What are the values of k and a ? (2 marks)

22. Trigonometry, 2ADV T3 EQ-Bank 4

The graph below has the equation $y = a\sin(bx) + c$ for $0 \leq x \leq 50$.



Determine the values of a, b and c . (3 marks)

23. Trigonometry, 2ADV T3 SM-Bank 12

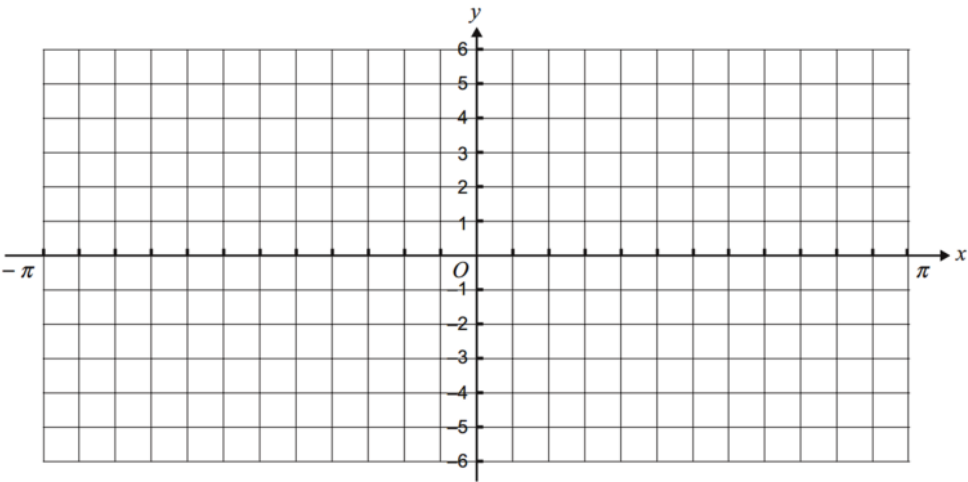
State the range and period of the function

$$h(x) = 4 + 3\cos\left(\frac{\pi x}{2}\right). \quad (2 \text{ marks})$$

24. Trigonometry, 2ADV T3 SM-Bank 14

For the function $f(x) = 5\cos\left(2\left(x + \frac{\pi}{3}\right)\right), \quad -\pi \leq x \leq \pi$

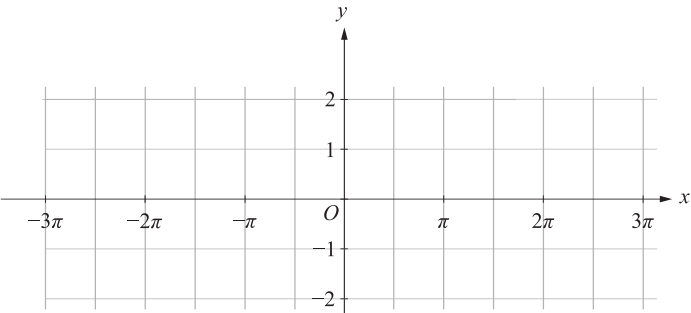
- a. Write down the amplitude and period of the function (2 marks)
- b. Sketch the graph of the function $f(x)$ on the set of axes below. Label axes intercepts with their coordinates. (3 marks)



25. Trigonometry, 2ADV T3 EQ-Bank 3

By drawing graphs on the number plane, show how many solutions exist for the equation

$$\cos x = \left| \frac{x - \pi}{4} \right| \text{ in the domain } (-\infty, \infty) \quad (3 \text{ marks})$$



26. Trigonometry, 2ADV T3 SM-Bank 8

$f(x) = 2\sin(2x)$ is defined in the domain $\left\{x: \frac{\pi}{8} \leq x < \frac{\pi}{3}\right\}$

What is the range of the function $f(x)$? (2 marks)

Worked Solutions

1. Trigonometry, 2ADV T3 2020 HSC 6 MC

$$-1 \leq \cos 3x \leq 1$$

$$-2 \leq 2\cos 3x \leq 2$$

$$3 \leq 5 + 2\cos 3x \leq 7$$

$$\therefore \text{Range } [3, 7]$$

$$\Rightarrow B$$

2. Trigonometry, 2ADV T3 SM-Bank 8 MC

$$\text{Graph centres around } y = 1$$

$$\text{Amplitude} = 3$$

$$\text{Period} = 4$$

$$\Rightarrow D$$

3. Trigonometry, 2ADV T3 SM-Bank 1 MC

$$\text{Range: } [-3 - 2, -3 + 2]$$

$$= [-5, -1]$$

$$\text{Period} = \frac{2\pi}{n} = \frac{2\pi}{3}$$

$$\Rightarrow A$$

4. Trigonometry, 2ADV T3 SM-Bank 2 MC

$$\text{Period} = \frac{2\pi}{n} = \frac{2\pi}{\frac{\pi}{2}} = 4$$

$$\text{Amplitude} = 2$$

$$\text{Graph centre line (median): } y = 1.$$

$$\therefore \text{Range} = [1 - 2, 1 + 2]$$

$$= [-1, 3]$$

$$\Rightarrow B$$

5. Trigonometry, 2ADV T3 SM-Bank 3 MC

$$\text{Period} = \frac{2\pi}{2} = \pi$$

$$\text{Range} = [-1 - 5, -1 + 5]$$

$$= [-6, 4]$$

$$\Rightarrow C$$

6. Trigonometry, 2ADV T3 SM-Bank 18 MC

$$n = \frac{\pi}{2}$$

$$\text{Period} = \frac{\pi}{n} = \frac{\pi}{\frac{\pi}{2}} = 2$$

$$\Rightarrow A$$

7. Trigonometry, 2ADV T3 SM-Bank 4 MC

$$\text{Period} = \frac{\pi}{2}$$

$$\Rightarrow \text{must be } C \text{ or } D$$

$$\text{Shift } y = \tan(x) \text{ right } \frac{\pi}{4}.$$

$$\Rightarrow C$$

8. Trigonometry, 2ADV T3 SM-Bank 6 MC

$$\begin{aligned} \text{Period} &= \frac{\pi}{n} \\ &= \frac{\pi}{\frac{1}{3}} \\ &= 3\pi \end{aligned}$$

$$\Rightarrow D$$

9. Trigonometry, 2ADV T3 SM-Bank 7 MC

$$\text{Amplitude} = 2 \quad (\text{range from } -1 \text{ to } 3)$$

$$\text{Graph centre line (median): } y = 1$$

$$\therefore \text{Eliminate } C \text{ and } D.$$

$$\text{Period} = \frac{2\pi}{3} - \frac{\pi}{6} = \frac{\pi}{2} \quad (\text{from graph})$$

$$\text{Consider option } B,$$

$$\text{Period} = \frac{2\pi}{n} = \frac{2\pi}{4} = \frac{\pi}{2}$$

$$\Rightarrow B$$

10. Trigonometry, 2ADV T3 2019 HSC 7 MC

$$a = \frac{1}{2}(5.5 - 2.5) = 1.5$$

$$\text{Since graph passes through } \left(\frac{\pi}{4}, 5.5\right):$$

$$5.5 = 1.5\sin\left(b \times \frac{\pi}{4}\right) + 4$$

$$\sin\left(b \times \frac{\pi}{4}\right) = 1$$

$$b \times \frac{\pi}{4} = \frac{\pi}{2}$$

$$\therefore b = 2$$

$$\Rightarrow D$$

11. Trigonometry, 2ADV T3 2013 HSC 6 MC

$$\text{At } x = 0, y = \sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \text{It cannot be A or C}$$

$$\text{Find } x \text{ when } y = 0,$$

$$\sin\left(2x + \frac{\pi}{3}\right) = 0$$

$$\therefore 2x + \frac{\pi}{3} = 0 \quad (\sin 0 = 0)$$

$$2x = -\frac{\pi}{3}$$

$$x = -\frac{\pi}{6}$$

$$\Rightarrow D$$

♦♦ Mean mark 34%

12. Trigonometry, 2ADV T3 2016 HSC 6 MC

$$\text{Period} = \frac{\pi}{n}$$

$$= \frac{\pi}{3}$$

$$\Rightarrow A$$

♦ Mean mark 42%.

13. Trigonometry, 2ADV T3 EQ-Bank 5

$$g(x) = \frac{1}{3}\sin(4x)$$

$$\Rightarrow \text{The new amplitude is one third of the original amplitude.}$$

$$\text{Period} = \frac{2\pi}{n} \Rightarrow n = \frac{1}{4}$$

$$\Rightarrow \text{The new period is one quarter of the original period.}$$

14. Trigonometry, 2ADV T3 2010 HSC 8c

i. $A = 4$

ii. Since the graph passes through $\left(\frac{\pi}{4}, 4\right)$

Substituting into $y = 4\sin bx$

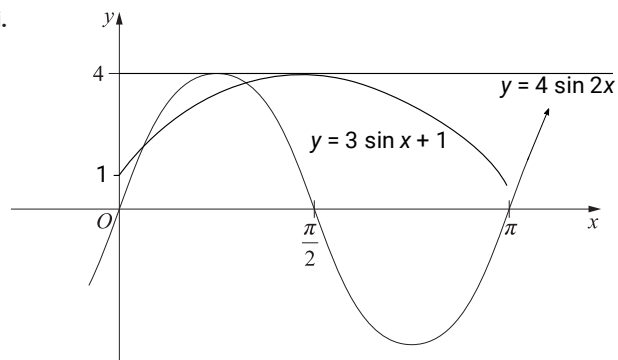
$$4\sin\left(b \times \frac{\pi}{4}\right) = 4$$

$$\sin\left(b \times \frac{\pi}{4}\right) = 1$$

$$b \times \frac{\pi}{4} = \frac{\pi}{2}$$

$$\therefore b = 2$$

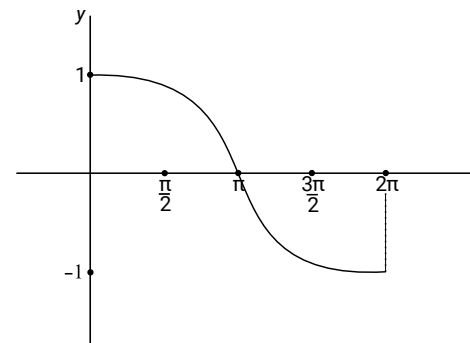
iii.



MARKER'S COMMENT: Graphs are consistently drawn too small by many students. Aim to make your diagrams 1/3 to 1/2 of a page.

15. Trigonometry, 2ADV T3 EQ-Bank 6

a.



b. $y = \cos\left(\frac{x}{2}\right)$

$$0.5 = \cos\left(\frac{x}{2}\right)$$

$$\frac{x}{2} = \cos^{-1}(0.5)$$

$$\frac{x}{2} = \frac{\pi}{3}, \frac{5\pi}{3}$$

$$x = \frac{2\pi}{3} \quad (0 \leq x \leq 2\pi)$$

16. Trigonometry, 2ADV T3 SM-Bank 9

i. $2\cos(x) + 1 = 0$

$$\cos(x) = -\frac{1}{2}$$

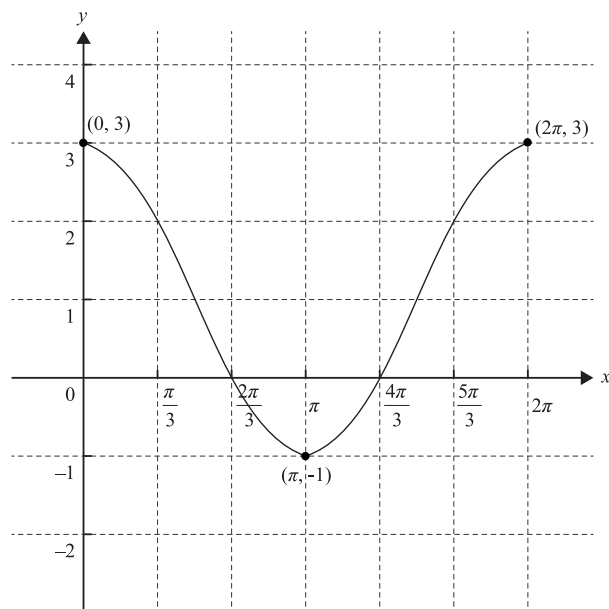
$$\Rightarrow \cos \frac{\pi}{3} = \frac{1}{2} \text{ and } \cos \text{ is negative}$$

in 2nd/3rd quadrant

$$\therefore x = \pi - \frac{\pi}{3}, \pi + \frac{\pi}{3}$$

$$= \frac{2\pi}{3}, \frac{4\pi}{3}$$

ii.



17. Trigonometry, 2ADV T3 2006 HSC 7b

i. $f(x) = 1 + 2\cos x$

$f(x)$ cuts the x -axis when $f(x) = 0$

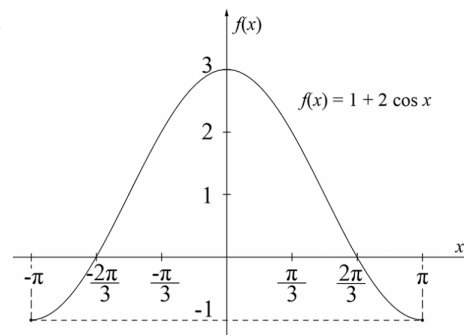
$$1 + 2\cos x = 0$$

$$2\cos x = -1$$

$$\cos x = -\frac{1}{2}$$

$$\therefore x = \frac{2\pi}{3} \dots \text{as required}$$

ii.



iii. Area = $\int_{-\frac{\pi}{2}}^{\frac{2\pi}{3}} 1 + 2\cos x \, dx$

$$= [x + 2\sin x]_{-\frac{\pi}{2}}^{\frac{2\pi}{3}}$$

$$= \left[\left(\frac{2\pi}{3} + 2\sin \frac{2\pi}{3} \right) - \left(-\frac{\pi}{2} + 2\sin \frac{-\pi}{2} \right) \right]$$

$$= \left(\frac{2\pi}{3} + 2 \times \frac{\sqrt{3}}{2} \right) - \left(-\frac{\pi}{2} + 2(-1) \right)$$

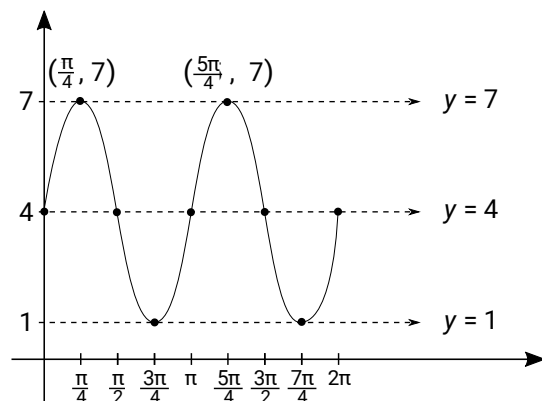
$$= \frac{2\pi}{3} + \sqrt{3} + \frac{\pi}{2} + 2$$

$$= \left(\frac{7\pi}{6} + \sqrt{3} + 2 \right) \text{ u}^2$$

18. Trigonometry, 2ADV T3 2017 HSC 14a

$$y = 4 + 3\sin 2x$$

$\Rightarrow$ Amplitude of 3 about $y = 4$



19. Trigonometry, 2ADV T3 2010 MET1 3

$$y = \tan(ax + b)$$

Substitute $(1, \sqrt{3}), (-1, -1)$ into equation:

$$\tan(a + b) = \sqrt{3}$$

$$\tan(b - a) = -1$$

$$a + b = \frac{\pi}{3} \dots (1)$$

$$b - a = -\frac{\pi}{4} \dots (2)$$

Add (1) + (2):

$$2b = \frac{\pi}{3} - \frac{\pi}{4}$$

$$b = \frac{\pi}{24}$$

Substitute into (1):

$$a + \frac{\pi}{24} = \frac{\pi}{3}$$

$$a = \frac{7\pi}{24}$$

20. Trigonometry, 2ADV T3 2021 HSC 20

Find x such that:

$$2\sin 4x = 1$$

$$\sin 4x = \frac{1}{2}$$

$$4x = \sin^{-1} \frac{1}{2}$$

$$4x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}, \dots$$

$$\therefore x = \frac{\pi}{24}, \frac{5\pi}{24} \quad \left(0 \leq x \leq \frac{\pi}{4}\right)$$

21. Trigonometry, 2ADV T3 2022 HSC 14

Amplitude = 4

$$\Rightarrow k = 4$$

Period = 6π

$$\frac{2\pi}{a} = 6\pi$$

$$6\pi a = 2\pi$$

$$\Rightarrow a = \frac{1}{3}$$

22. Trigonometry, 2ADV T3 EQ-Bank 4

$$\text{Amplitude} = \frac{40 - 8}{2} = 16 \Rightarrow a = 16$$

$$\text{Centre of motion} = 24 \Rightarrow c = 24$$

$$\text{Period} = \frac{360}{n} = 40 \Rightarrow n = b = 9$$

23. Trigonometry, 2ADV T3 SM-Bank 12

$$-1 \leq \cos\left(\frac{\pi x}{2}\right) \leq 1$$

$$-3 \leq 3\cos\left(\frac{\pi x}{2}\right) \leq 3$$

$$1 \leq 4 + 3\cos\left(\frac{\pi x}{2}\right) \leq 7$$

$$\therefore \text{Range: } 1 \leq y \leq 7$$

$$\text{Period} = \frac{2\pi}{n} = \frac{2\pi}{\frac{\pi}{2}} = 4$$

24. Trigonometry, 2ADV T3 SM-Bank 14

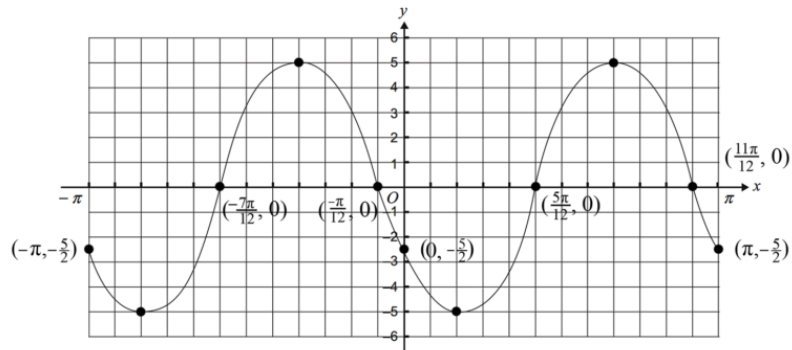
a. Amplitude = 5

$$\text{Period} = \frac{2\pi}{2} = \pi$$

b. Shift $y = 5\cos(2x)$ left $\frac{\pi}{3}$ units.

$$\text{Period} = \pi$$

Endpoints are $\left(-\pi, -\frac{5}{2}\right)$ and $\left(\pi, -\frac{5}{2}\right)$



25. Trigonometry, 2ADV T3 EQ-Bank 3

Sketch:

$$y = \cos x$$

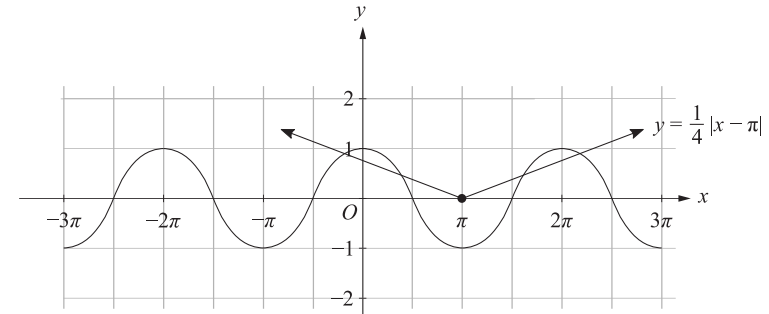
$$y = \left| \frac{x-\pi}{4} \right|$$

Translate π units to the right:

$$y = |x| \Rightarrow y = |x - \pi|$$

Multiply by $\frac{1}{4}$:

$$y = |x - \pi| \Rightarrow y = \frac{1}{4}|x - \pi| = \left| \frac{x - \pi}{4} \right|$$



$\therefore$ There are 4 solutions.

26. Trigonometry, 2ADV T3 SM-Bank 8

$\sin(2x)_{\max}$ occurs when $x = \frac{\pi}{4}$ (within domain)

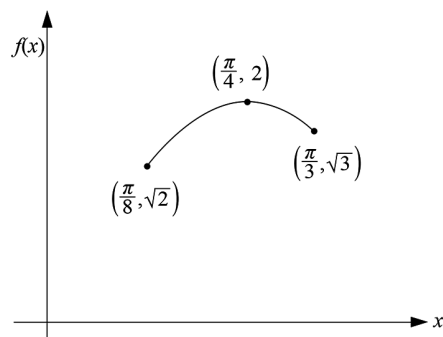
♦ Mean mark 45%.

$$\Rightarrow f(x)_{\max} = 2\sin\left(\frac{\pi}{2}\right) = 2$$

Checking endpoints:

When $x = \frac{\pi}{8} \Rightarrow y = 2\sin\left(\frac{\pi}{4}\right) = \sqrt{2}$

When $x = \frac{\pi}{3} \Rightarrow y = 2\sin\left(\frac{2\pi}{3}\right) = \sqrt{3}$



$$\therefore \text{Range} = [\sqrt{2}, 2],$$